NUCLEAR FUEL CYCLE

International Nuclear Waste Disposal Concepts

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- There have been several proposals for regional and international repositories for disposal of high-level nuclear waste, and in 2003 the concept received strong endorsement from the head of the International Atomic Energy Agency.
- The European Commission is funding studies to assess the feasibility of European regional waste repositories for countries with relatively little nuclear waste.
- Arising from these studies, 14 EU countries resolved to set up the European Repository Development Organisation (ERDO) to collaborate on nuclear waste disposal.
- Since 2009 the International Framework for Nuclear Energy Cooperation (IFNEC) has focused on possible multinational repositories for their safety, security and environmental benefits.
- In May 2016 a high-level commission in South Australia recommended the establishment of an international repository there.
- In 2019 IFNEC suggested that it was not timely to develop a multinational repository until national repository projects were more advanced.

Privacy - Terms

Responsibility for waste

At present there is clear and unequivocal understanding that each country is ethically and legally responsible for its own waste, therefore the default position is that all nuclear waste will be disposed of in each of the 50 or so countries concerned.

The main ingredients of high-level nuclear waste are created in the nuclear reactors which make the electricity in 31 countries. They are essentially not left-overs from imported uranium. There is thus no moral obligation on uranium suppliers in respect to the waste, other than that involved in safeguards procedures.

For instance, Australian uranium is supplied under safeguards, which are essentially accounting and inspection procedures to ensure that neither the uranium nor any product of it (e.g. plutonium) contribute to fulfilling the aspirations of anyone wanting to build weapons. With the International Atomic Energy Agency (IAEA), the Australian Safeguards & Non-Proliferation Office tracks 'Australian-obligated nuclear materials' (AONM) all the way through to used fuel, reprocessing (if undertaken), and recycling of plutonium (if separated) in mixed oxide fuel. The same kind of arrangements apply to Canadian uranium.

Thus any international waste repository has implications under the Nuclear Non-Proliferation Treaty (NPT). The trustworthiness and standing of the host country is fundamental to the project's acceptability to NPT states, which comprise virtually every country but India, Pakistan, Israel and North Korea. Also, the international treaty produced by the IAEA and signed by most nations of the world in 1997 covering the management and disposal of used fuel and high-level waste requires that the host facility or system meets the highest national and international standards.

Even countries such as Australia with no nuclear power have a need for secure disposal of long-lived radioactive waste from their research reactors.

International repositories

In November 2003, Mohamed ElBaradei, Director-General of the UN's International Atomic Energy Agency (IAEA), said to the UN General Assembly: "We should ... consider multinational approaches to the management and disposal of spent fuel and radioactive waste. Over 50 countries currently have spent fuel stored in temporary locations, awaiting reprocessing or disposal. Not all countries have the appropriate geological conditions for such disposal – and, for many countries with small nuclear programmes, the financial and human resources required for the construction and operation of a geological disposal facility are daunting."

In an October article he included research reactors in the scope of this suggestion and concluded: "Considerable advantages – in cost, safety, security and non-proliferation – would be gained from international cooperation in these stages of the nuclear fuel cycle."

Nearly 25 years earlier, in 1980, the IAEA-sponsored International Nuclear Fuel Cycle Evaluation (INFCE) waste management and disposal report had firmly recommended that proposals "for establishing multinational and international repositories should be elaborated" due to their non-proliferation advantages. In addition, it said that centralized facilities for disposal of spent fuel and/or vitrified high-level waste could "reduce the diversion risk" and be more economical.

Individual waste repositories for spent nuclear fuel and other high-level waste need to be reliably secure. Achieving high security means:

- They can make a vital contribution to global environmental safety by ensuring that radioactive substances are permanently removed from the human environment.
- They can greatly enhance global security in the broader sense by preventing the malicious use of fissile and radiological materials.

Insofar as these functions are less than fully assured in any of the 50 countries concerned with radioactive waste, there is a justification for some kind of international collaboration and facilities, possibly on a regional basis. In particular, the second point is arguably best achieved by international collaboration under IAEA auspices. McCombie and Chapman (2004)[1] have presented arguments that the international community should take concrete steps to expedite credible proposals for shared repositories.

In 2004 the IAEA put forward three concepts for multinational repositories:

- Incremental addition to a large national programme.
- A supranational facility with international management and control.
- Collaborative partnering among countries for a multinational repository.

While most countries should be able to find suitably safe sites in stable geological formations, demonstrating this safety so as to create public confidence is best achieved where there is simple geology.

Certainly, there is a broad consensus that geological disposal is the only way of ensuring adequate safety and security in the long-term management of used fuel which is treated as waste, and also of separated high-level radioactive waste which, if recycling of actinides becomes established, eventually will mainly be fission products.

While acknowledging each county's responsibility for its own waste, the limits to the logic on indigenous disposal can be seen from the changing national borders within Europe over the last century. For example, in the case of Slovenia (which has one nuclear power reactor), its capital city Ljubljana has politically lain within seven different states in the last 100 years.

The May 2016 South Australian Nuclear Fuel Cycle Royal Commission report[2] notes:

There are international models that address the transfer of waste between countries. The Basel Convention, which applies to hazardous wastes other than radioactive waste, imposes requirements upon the transfer of hazardous wastes between countries; namely the transfer shall only take place where prior

informed consent has been received and only if the transfer represents an environmentally sound solution. Hazardous wastes are commercially transferred under this regime. While the Joint Convention applies equivalent requirements to transfers of radioactive waste between countries, there are no operating models for the commercial transfer of used fuel for disposal.

Various organizations have looked into potential concepts. There are, however, commercial models for the transfer of used fuel between countries for reprocessing, as well as the take-back of fuel from reactors built by Rosatom, the Russian state nuclear corporation... The United Kingdom has reprocessed used fuel for many countries but does not accept the waste products for disposal. In all cases, transfers can only take place if the recipient country has the capacity to manage the waste safely and where such transfer has been agreed between the countries concerned.

In October 2016 the Reliable Nuclear Fuel Services Working Group (RNFSWG) of the International Framework for Nuclear Energy Cooperation (IFNEC) released a report, Practical Considerations to Begin Resolving the Final Spent Fuel Disposal Pathway for Countries with Small Nuclear Programs, based on previous work initiated by Arius as well as studies done by the IAEA and the OECD Nuclear Energy Agency. It notes that under the IAEA Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management, contracting parties have agreed that the country that discharges the spent fuel and receives the benefits of the power generated bears the responsibility for its management, including disposal. Since disposal is a relatively major burden with small quantities, due to high fixed costs, some international collaboration is suggested. This is the dual track beyond simply national approaches, allowing a country to offer repository facilities as a service provider.

This was fleshed out in an IFNEC presentation to the IAEA in mid-2019, Development of Multinational Repository Concept: Exploring Alternative Approaches to Financing Multinational Repository. This reported on an IFNEC workshop held in Paris in 2018 which considered the spending profile on a repository from siting and licensing through construction (35% of total), operation (45% of total) and closure. It concluded however that multinational repository development was not yet timely, though the concept had potential and should continue to be studied. Countries with small nuclear programmes should set aside funds in anticipation. IFNEC should continue to work on linking any potential service provider with potential customer countries.

South Australian proposal

In May 2016 the South Australian Nuclear Fuel Cycle Royal Commission reported. A major recommendation was that a facility for the disposal of international used nuclear fuel and intermediate-level waste should be established. It found that the state "has the necessary attributes and capabilities to develop a world-class waste disposal facility, and to do so safely." Based on a "cautious and conservative approach", from assessments of used fuel inventories and potential global interest the commission determined that such a facility could generate more than A$100 billion in income in excess of expenditure (including a reserve fund of A$32 billion for facility closure and ongoing monitoring) over the 120-year life of the project.

The World Nuclear Association said that the report had "fundamentally changed the nature of the global nuclear waste discourse," and a multinational waste facility based in South Australia would provide a welcome option for countries operating nuclear facilities today. It would be a “viable alternative" to national projects. Such a large multinational waste storage facility would be a world first and should offer advantages in terms of siting and economics when compared to smaller national approaches.[3]

The South Australian Royal Commission into the Nuclear Fuel Cycle report (2016) continues:

Under the Joint Convention [on the Safety of Spent Fuel Management and the Safety of Radioactive Waste Management], any proposal to store and dispose of used fuel in South Australia would require agreement between the countries concerned. In Australia, treaty level agreements would need to be developed between the federal government and the relevant overseas government. An agreement would also need to specify arrangements between the Australian Government and the Government of South Australia, to ensure these commitments were fulfilled. Further agreements may be required with third party countries: for example, if they have supplied uranium to the country wishing to store and dispose of used fuel in South Australia...

As Australia is a net exporter of energy, it has a significant role to play in assisting other countries to lower their carbon emissions. This includes countries with less opportunity for large-scale renewable energy deployment than Australia, for whom nuclear power makes a substantial contribution to their production of low-carbon energy. For new nuclear entrants or countries with little

prospect of siting their own used fuel disposal facilities, an international solution would remove a significant impediment to the new or ongoing use of nuclear power as a low-carbon technology. As a result, Australia would derive a reputational and financial benefit by hosting a facility for the disposal of international used fuel.

Excluding countries which are committed to developing national solutions or already have structured programmes leading to a domestic facility, and those with laws prohibiting export of waste, the report identifies 90,000 tonnes of used fuel and almost 270,000 cubic metres of intermediate-levels waste (ILW) as a potentially accessible market and available for an Australian repository today. These amounts, comprising about one-quarter of actual world used fuel and ILW, are increasing annually. By 2090 the amounts are 276,500 tonnes and 782,000 m^3 respectively. The viability analysis is based on half these amounts.

Based on detailed analysis, the commission considered that a reasonable baseline price for the purpose of assessing viability would be A$1.75 million/tHM for used fuel. This is based on a reasonable baseline 'willingness to pay' estimate of A$1.95 million/tHM, less A$0.2 million/tHM to account for costs incurred by customers in preparing and delivering the waste to South Australia.

Regarding intermediate-level waste, the Royal Commission report says:

The management and disposal of intermediate-level waste commands a far lower willingness to pay than for used fuel. This is due to a country's ability to stockpile intermediate-level waste arising from nuclear power plants or other sources (such as decommissioned nuclear facilities) within shielded containers far more readily than used fuel...

However, a 2011 report from the UK Department of Energy and Climate Change has suggested that £25,900 per m^3 (in current terms, A$66,000 per m^3) represents a levy that ought be imposed on nuclear power plant operators to reflect current costs and the potential for future increases. In the interests of conservatism, and to address the costs of packaging and transport (which are not as well defined as for used fuel) a price to charge of A$40,000 per m^3 is considered appropriate for the purposes of a viability analysis.

Interim above-ground storage for used fuel inside heavily engineered, purpose-built casks would be followed by deep geological disposal, the repository design based on that under construction at Olkiluoto in Finland at 400-450 metres depth. In the

analysis, the geological disposal facility for used fuel is notionally co-located with an intermediate-level waste facility, where those packages are placed in medium-depth vaults of 50-250 metres. The report said that integrated facilities with capacity to store and dispose of used fuel would be viable.

On a number of realistic scenarios, such a facility would be highly profitable. The timeline for establishing an interim storage facility and associated transport infrastructure, including harbour, port and railway would be 11 years after project commencement. Transferring used fuel and ILW from surface storage to underground repository would begin at 28 years. Revenue would thus commence at year 11, based on 3000 t/yr used fuel, so A$5.25 billion/yr over 30 years from then, simply for the fuel. Total capital cost would be A$41 billion over some years including a 25% contingency, and operating expenses from year 11 about A$0.878 million (in 2015 $). Overall total revenue (in undiscounted terms) would be more than A$257 billion, with total costs of A$145 billion. The project would add 4.7% to gross state product and 1.9% to employment by 2030.

Legislative changes would be necessary at state and federal levels.

ARIUS and Europe – ERDO

Early in 2002 a new, non-commercial body to promote the concept of regional and international facilities for storage and disposal of all types of long-lived nuclear waste was set up. This is ARIUS – the Association for Regional and International Underground Storage. A key objective is to explore ways of providing shared radioactive waste management approaches and facilities, in particular storage and disposal facilities for smaller users. Membership is open and comprises countries with small nuclear programmes as well as industrial organizations with relevant interests. Arius is a successor to the Pangea proposal (see below).

Arius focused initially on Europe, and the feasibility of regional repositories there. A 2003 European Commission proposed directive said that geological disposal of radioactive waste was preferred and: "An approach involving two or more countries could also offer advantages especially to countries that have no or limited nuclear programmes, insofar as it would provide a safe and less costly solution for all parties involved."[4]

In mid-2003 Arius initiated the SAPIERR project (Support Action: Pilot Initiative for European Regional Repositories), which obtained

European Commission approval. This was undertaken over two years to 2005 to help the EC grapple with the regional repository issue as flagged earlier in the proposed EC radioactive waste directive. It allowed potential options for regional collaboration and for regional repositories to be identified, though it did not extend to site identification. Slovakia provided the project coordination.

Following this pilot study, in September 2006 the EC-funded SAPIERR II (Strategic Action Plan for Implementation of European Regional Repositories) project to assess the feasibility of European regional waste repositories was commenced, indicating a recognition in the EU that implementing 25 national repositories is not optimal economically or for safety and security. The project was in line with proposals from the International Atomic Energy Agency (IAEA), Russia and the USA (with the Global Nuclear Energy Partnership, now IFNEC) for multilateral cooperation in the fuel cycle in order to enhance global security. Shared repositories for high-level nuclear waste are an important element of this.

The SAPIERR II project held its final symposium January 2009 in Brussels. The results of studies on the viability of shared, regional European geological repositories were presented to 50 participants from 21 countries. The aspects considered included organizational and legal issues, economic impacts, safety and security considerations, and public and political attitudes to multinational repositories.

The main outcome of this was that 14 countries* resolved to move towards setting up the European Repository Development Organisation (ERDO). The first step in the strategy was the establishment of a self-financing working group (ERDO-WG) of interested countries in 2009 to prepare a consensus model to be agreed for ERDO, using the SAPIERR II findings as a starting point. It proposes a sister organization to waste agencies from European countries that have opted for a purely national repository programme. This model was presented to potentially interested countries, and internal discussions are still in progress in several of these on adopting the multinational solution as one option when responding to the requirements placed on EU member states by the EC Waste Directive of 2011.[a] Some member states have already decided to follow this course. The ERDO-WG secretariat is provided by ARIUS and the administration by the Netherlands waste agency, COVRA.

* Austria, Bulgaria, Czech Republic, Denmark, Estonia, Ireland, Italy, Latvia, Lithuania, Netherlands, Poland, Romania, Slovakia, and Slovenia, though some subsequently did not continue in the

ongoing ERDO-WG. In April 2020 the WG active membership comprised Austria, Croatia, Denmark, Italy, Netherlands, Norway, Poland and Slovenia.

In September 2014 Arius together with European partners submitted the COMS-WD (cooperation between member states responding to the Waste Directive) proposal to the €80 billion Horizon 2020 EU research and innovation programme. Nine EU countries (Austria, Croatia, Denmark, Greece, Italy, Netherlands, Poland, Portugal, and Slovenia) and the Joint Research Centre Institute for Transuranium Elements (JRC-ITU) were involved with Arius in this. JRC-ITU involvement is to ensure continued cooperation with other EU projects. However, the proposal was rejected as being 'out of scope' for Horizon 2020. Nevertheless, the multinational or dual-track option is being included by some countries in their strategies responding to the Waste Directive – Netherlands and Slovenia so far.

ARIUS and the Persian Gulf

Alongside ERDO, Arius is evaluating whether similar, regional shared solutions would be appropriate for and of interest to emerging nuclear power programmes in the Persian Gulf, Middle East and North Africa (MENA) region and also southeast Asia. The overall aim is to assess the interest within each region of working towards regional repository development organizations (RDOs) similar to the Europe's ERDO. A scoping study was completed in 2011 and funding for a further two-year project was awarded in mid-2011 by two US foundations. In 2011 the United Arab Emirates' (UAE's) permanent representative to the IAEA raised the possibility of a regional repository and said: "Among GCC countries there is potential for a lot of cooperation in this area, including waste management and waste disposal." A final project report from the scoping study was in February 2016.

In April 2012, an initial meeting took place in Abu Dhabi, hosted by the Federal Authority for Nuclear Regulation (FANR) of the UAE and supported by the IAEA. Around 35 participants attended, primarily from UAE waste management and planning organizations. The UAE was an obvious host for a first meeting since it has the most advanced nuclear power programme in the region and has also formally committed to a dual track radioactive waste management strategy that involves developing a national storage and disposal programme in parallel with exploring regional cooperation options. A follow-up to the Abu Dhabi meeting was in November 2012 in

Tunisia, hosted by the Arab Atomic Energy Agency (AAEA), with a widened group of participating MENA countries. In the six countries comprising the Gulf Cooperation Council region, which includes two nations with expanding nuclear power programmes (the UAE and Saudi Arabia), consideration is also being given to launching a joint project on the feasibility of shared storage and/or disposal facilities. Further discussion has followed, including at a January 2015 workshop in Abu Dhabi.

Arius estimated that the cost of such a shared regional repository in the Middle East might be $4 billion, but would offer a large payback in the form of regional security. It would not be needed before about 2080.

The European and MENA initiatives will act as role models for a possible southeast Asian initiative, where interest has been shown by representatives of potential nuclear power newcomer nations such as Malaysia and Vietnam and where involvement with the mature nuclear power programmes of the region (e.g. in South Korea and on Taiwan) will add an extra dimension to the issue. Following a January 2014 meeting in Indonesia, further meetings are envisaged.

An EU group, the Implementing of Geological Disposal of Radwaste Technology Platform (IGD-TP), was established in 2009 by several EU organizations to support research related to its vision and assist the various EU programmes. It does not have government-level representation but is partly funded by the EU.

Fuel leasing

There have been a number of proposals for fuel leasing. At present the normal process is that a utility buys uranium from a mining company, then gets it converted, enriched and fabricated, before using it and discharging the used fuel. That used fuel becomes the responsibility of the country in which it has been used, as outlined in the opening paragraphs at the beginning of this page.

Fuel leasing is an alternative concept whereby the utility leases its fabricated fuel from a supplier, probably in another country, and after it has been used that supplier takes it back. This concept is not yet in use except for some very limited applications, mainly for Russian-built nuclear power plants in NPT non-weapons states (e.g. Bangladesh, Turkey, Iran). The supplier would then add the leased used fuel to its own larger stocks to be stored for later

disposal or reprocessing and recycling, in which case the valuable components would belong to the fuel supplier/leaser.

The concept was part of the Global Nuclear Energy Partnership (now IFNEC) launched in 2006, in order to restrict the spread of sensitive technologies such as enrichment, but it did not come to fruition under that programme. Of all the nuclear suppliers, Russia has expressed the most support for fuel leasing and take-back. Russia's fuel supply contract with Iran involves leasing, and Iran is required to send the used fuel back to Russia. Russian law allows the import of foreign used fuel for reprocessing and the return of waste (as with reprocessing other countries' used fuel in France and the UK), and Russia's default position in supplying reactors to NPT non-weapons states is to take back Russian-origin fuel without requiring return of the waste, as with Iran.

The Pangea proposal

In the 1990s, UK-based company Pangea Resources identified Australia, southern Africa, Argentina and western China as having the most appropriate geological credentials for a deep geologic repository, with Australia being favoured on economic and political grounds. It would be located where the geology has been stable for several hundred million years, so that there need not be total reliance on a robust engineered barrier system to keep the waste securely isolated for thousands of years.

It would be a commercial undertaking and would have dedicated port and rail infrastructure. It would take spent fuel and other waste from commercial reactors, and possibly also material from weapons disposal programmes.

Pangea summed up the situation thus:[6]

By taking a fresh look at the reasons for the difficulties which have faced most national repository programs, and discarding the preconception that each country must develop its own disposal facilities, it is possible to define a class of simple, superior high-isolation sites which may provide a multinational basis for solving the nuclear waste disposal problem.

The relatively small volumes of high-level wastes or spent fuel which arise from nuclear power production make shared repositories a feasible proposition. For small countries, the economies of scale which can be achieved make the concept attractive. For all countries, objective consideration of the relative

merits of national and multinational solutions is a prudent part of planning the management of long-lived radioactive wastes.

Early in 1999 Pangea Resources released its project proposal to the Australian public, expecting this "to initiate discussions which will enable us to more fully assess the feasibility and strategy of our proposal ... on (its) merits."

The initial response from the federal government however was to reiterate Australia's long-standing and bipartisan policy of not importing nuclear wastes and saying that there was no immediate intention of considering such a proposal. Then, after only cursory consideration, the Western Australian parliament passed a bill to make it illegal to dispose of foreign high-level waste in the state without specific parliamentary approval. Pangea continued its geological investigations in that state while extending its feasibility study to other potential host regions.

Pangea's objectives were to site a deep geologic disposal facility in a region where the geology and biosphere conditions meet the test of simplicity coupled with robustness. This is required to demonstrate that the performance of the facility from a safety standpoint will meet the highest international standards and international safeguard requirements. In addition to its ideal geological characteristics, the host country should preferably be a first-world, stable democracy, familiar with high-technology enterprises.

Pangea's strategy implied that the geological barrier can be the primary safety barrier, in contrast to some other potential repository concepts where the waste form and the engineered barriers are required to be more dominant. There is a side benefit in that less complex and less expensive engineered barriers may be sufficient. Pangea also saw a potential public acceptance benefit, in that reliance on a simple geological barrier might be more readily understood and accepted.

The Pangea concept envisaged a dedicated port and rail link to an inland repository site covering perhaps 5 km^2 on the surface and 20 km^2 underground (500 metres down). There would be a fleet of 35 dedicated and purpose-built ships at any one time.

Australian antecedents

The Pangea proposal had Australian antecedents. For instance, in 1983 the Labor government commissioned a report from the

Australian Science and Technology Council (ASTEC) on Australia's role in the nuclear fuel cycle. The 1984 ASTEC report[7] recommended not only proceeding with uranium mining, but also becoming involved with other stages of the fuel cycle such as enrichment. It also flagged the "particular need for international collaboration in developing (high-level) waste management programs" and the desirability of enabling access to the highest quality geological sites for disposal of high-level waste (paragraph 2.3.51).

The Pangea concept can be traced to the Synroc Study Group, which began its activities in late December 1988. The Synroc Study Group was a vehicle set up by the Australian government to study the commercial potential for Synroc in a global context. It was conducted by four leading Australian resource companies, assisted by the Australian Nuclear Science & Technology Organisation (ANSTO) and the Research School of Earth Sciences at the Australian National University.

It is not current government policy to import nuclear waste into Australia, and this has been reiterated several times by government sources. In order for any repository proposal to be successful in any country, it would have to receive various government permits and meet full-scale licensing and environmental impact requirements in its host country.

In the event, the proposal proved to be ahead of its time for Australia, and apparently elsewhere.

Russian plans 2001-06

In 2001 the Russian parliament (Duma) passed legislation to allow the import of spent nuclear fuel. The President signed this into law and set up a special commission to approve and oversee such imports. The commission would have 20 members, five each from the Duma, the Council (upper house), the government and presidential nominees. It would be chaired by Zhores Alferoy, a parliamentarian who is also Vice President of the Russian Academy of Sciences and a Nobel Prize physicist.

In 2003 Krasnokamensk was suggested as the site for a major spent fuel repository – it is a city 7000 km east of Moscow in the Chita region and is a centre for uranium mining and milling, the mines being run by the Priargunsky Combine.

This scheme was progressed in 2005 when the Duma ratified the Vienna Convention on Civil Liability for Nuclear Damage. However in July 2006 Rosatom announced it would not proceed with taking any foreign-origin used fuel.

Notes & references

Notes

a. At the end of 2011, the ERDO Working Group, led by representatives of the Dutch and Slovakian national programmes, prepared a set of documents outlining the possible structure, operating methods and financing options for a formal, multinational, European waste management agency (ERDO). These documents were submitted to all EU governments that might be interested in actively participating in the preparation of this next step, and also provided as information to those European countries, such as Sweden, Finland and France, that have decided on purely national approaches to managing their radioactive waste. Discussions with potential participants have been under way since 2012, and a number of positive responses received. In December 2013 the EC, the European Nuclear Energy Forum (ENEF) Waste Group and the ERDO-WG organized a meeting in Luxembourg to consider how to help all EU member states meet the requirements set by the EC Waste Directive of 2011.[5] The meeting addressed the fact that some EU member states are not well equipped to meet the directive's targets, but could develop a common programme for addressing many of its requirements. [Back]

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